



Microwave assisted biodiesel production from chicken feather meal oil using Bio-Nano Calcium oxide derived from chicken egg shell

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ABSTRACT

Environmental concerns have increased the search for greener measures to mitigate pollution issues. Bio Nano CaO was synthesized and reduced by tea decoction extracted from chicken egg shell using tea decoction. The synthesized material was characterized by physico-chemical techniques such as XRD, TGA, BET surface area analyser, TGA and SEM analyses. XRD analysis confirmed the crystalline nature of material. The prepared material was found to be stable till 100 °C from TGA study. The SEM pictures displayed uniform and discrete particles which portrays the high probable sites that maximises the catalytic activity. The optimization of microwave assisted Biodiesel synthesis from chicken oil through Transesterification process using the bio-synthesized catalytic material was the main aim of the study. A 500 W microwave irradiation of Chicken feather meal oil using 8:1 Methanol:oil input, 1% Bio Nano CaO concentration, 5 min of reaction time resulted in 95% conversion of chicken feather meal oil into chicken feather meal methyl esters. The Biodiesel was showed low viscosity (4.15 mm²/s), high heating value (50 MJ/kg), high flash point (153 °C), reasonable pour point (12 °C) and good cetane number (50 min). The future works will be concentrated on the engine studies related to Torque, fuel consumption, emission data by using the synthesized Biodiesel.

1. Introduction

The growth of automobiles is parallel to the world's steering population (Miglan, 2019). Specifically, developing countries like China and India are facing a human rate faces the worst scenario in air pollution due to the toxic air pollutants like SO_x, NO_x, aerosols and Particulate matter and the situation seems to worsen in due course (Kouravand and Karami, 2018). The problem gets bloated with a) Exhaustion of crude oil by continuous and rapid exploitation (Höök, 2009), b) Doubling petroleum prices and last but not the least (Ni et al.,

2020; Provornaya et al., 2020), c) Severe global warming effects caused by non-stop combustion of fossils (Choi and Manousiouthakis, 2020; Perera, 2018; Wang et al., 2020; Yang et al., 2020). Independence from dependency on petroleum derived fuel could be attained only by the introduction of liquid fuels having renewable bio resource base (Patel and Shah, 2021; Su et al., 2020). Triglycerides of non-edible vegetable oil, lipids of microorganisms, cellulose of energy crops are the highly concentrated feed stocks (Chatterjee and Mohan, 2018; de Oliveira et al., 2020; Passoth and Sandgren, 2019). Waste products i.e., red mud, chicken egg shell, animal tallow is the latest approach towards biodiesel

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synthesis. Chicken feather is the highly discarded part inspite of fat content (10%–12%) present in it. However, Fertilizers and animal feed are prepared from thermally processed Chicken feather meal (CFM). CFM contains 10–12% of non-edible fat and research works on converting the CFM lipids to liquid fuels (Rosales-Calderon and Arantes, 2019). Research work on converting the CFM lipids to liquid fuels is reported and has to be extensively studied in future. The process existing in the liquid fuel production field are;

- 1) **Catalytic cracking** – The feedstock is cracked at higher temperature (450 °C – 550 °C) with assistance of catalyst such as metal oxides and porous materials to yield liquid hydrocarbons (Cao et al., 2020). The attempt of vaporizing the oil results in carbonization of small fragments of oil which eventually clogs the pores of catalyst (Merchan-Merchan et al., 2018; Zhao et al., 2019).
- 2) **Fermentation** – Biological digestion of cellulose rich feeds by enzyme catalyst results in Bio alcohols (Saire-Saire et al., 2020). The technique is not affordable and up scaling the process becomes difficult. Besides, the fuel is ultra clean and has good calorific value (Wang et al., 2019).
- 3) **Pyrolysis** – Bio oil is the product of pyrolysis or combustion of oils or plastics at relatively very high temperate (650 °C). The pyrolyzed oil definitely needs to be upgraded for its utilization in engine, the fuel is highly oxygenated which lowers the heating value of the fuel (Din et al., 2020).
- 4) **Transesterification** - The fatty acid methyl esters prepared from oils by trans esterifying process in the presence of alcohol at comparatively low temperature than other processes is known as biodiesel (Li et al., 2020). The number of hydrocarbon chain of biodiesel (C18) resembles the diesel fraction and hence it is apt to term this fuel as biodiesel. Industrial production of biodiesel has also been started in the recent years owing to the a) simple technology, b) Large quantity, c) Comparable calorific value, d) Reasonable cost of operation and e) characteristics like environmental benign, renewable and biodegradable (Appavu et al., 2021; Saikia et al., 2019). The only shortcoming of this process is co-product of glycerine as a product and this makes the product recovery more prolonged. Heterogeneous solid base materials dominate the field due to their easy recovery, which is not require water wash treatment and excellent reusability (Sundya et al., 2020). The side product; Glycerin is also free from any trace of salt contaminants and hence this can be concluded as greener option. Nano heterogenous materials exhibit higher catalytic activity owing to their active acidic sites in large number which in turn increases their surface area (Gardy et al., 2019; Gajur et al., 2020). These unique features add credit by improving the rate of reaction in low duration and targeting specific products with uncountable catalysts that exist in practise, CaO is one of the catalysts as promising heterogeneous base catalyst due to its cost effectiveness as very less quantity of glycerol is obtained as side product. The oxygen anion present on the CaO surface is responsible for the catalytic impact of CaO in the transesterification reaction. CaO has an ionic crystal structure and is an alkaline earth oxide. The low Pauling scale value of 1.00 indicates the weak acidity of calcium cation as per Lewis theory. As a result, the conjugated oxygen anion has a powerful basic property. Prayitno et al. (2020), Degfie et al. (2019) and Habte et al. (2019) prepared CaO using egg shell as precursor and size reduction was done by thermal methods. The large consumption of chicken eggs makes their availability abundant and also the easy derivation of calcium oxide from the resource increases the research in it. The bio reduction of copper oxide and Calcium oxide is reported by Khatami et al. (2019), Sutrathar et al. (2014), Marquis et al. (2016) and Kiranmai et al. (2017). A decent conversion by the conventional transesterification technique will consume excess energy and require long reaction time (1 h–8 h) due to direct heating. Whereas, Microwave assisted biodiesel synthesis can be completed in shorter time with good

conversion rate as the energy is directly delivered to mixture instead of energy conduction from reactor vessel to mixture (Chipurici et al., 2019). Further, the separation process also feels less tedious compared to old process. Alcohol is highly activated under microwave irradiation with frequency from 0.3 to 300 GHz and hence best suited for μ wave assisted methods. The heat is generated in the vessel due to dipole rotation of alcohol and other ions of solution. In this article, chicken egg shell is the bio precursor to synthesize CaO catalyst. The size reduction of these materials were done using tea decoction. Biodiesel was produced using the synthesized material from CFM feedstock under microwave illumination. The Biodiesel was analysed using ASTM standards and the results are reported herein.

2. Materials and methods

The large canteen infrastructure of our college St. Joseph's Institute of Technology, consumes lot of chicken eggs. Chicken egg shell (CES) is massively neglected as was along with CES, large quantity of vegetable wastes is also discarded. Radish root was received from the college campus itself. The canteen serves non-vegetarian food almost every day and specifically chicken scrap of chicken feather is shelved. The tea dust from the college café was reused to extract the decoction. Therefore, radish, used tea dust and chicken feather was procured from our college canteen. The only chemical used for the synthesis was methanol and was applied by CDH, India.

2.1. Processing of precursors and feedstock

2.1.1. Preparation of CES and tea decoction

The dried egg shells were washed with water and dried in air oven at 100 °C. 100 mL of water was added to 50 g of used tea dust and heated at 100 °C. The solution was filtered and the extract was stored in sealed container below 25 °C.

2.1.2. Preparation of CFM

About 150 g of CFM was washed vigorously for several times to expel excrements and was heated at 100 °C in 1.5 L of distilled water upon continuous stirring at on a magnetic stirrer for 1 h. The mixture was cooled to room temperature and centrifuged at 7500 rpm (CLS430055, Sigma Aldrich) to separate fat from water. Fatty layer occupied the top layer and water was retained at the bottom. The free fatty acids (FFA) was removed from obtained fat by adding 20 mL of 0.5% KOH solution at 70–80 °C in the same centrifuge speed. A homogeneous solution was prepared by heating the fat content for 25 min at 45 °C to 50 °C.

2.1.3. Preparation of CaO from CES and reduction using tea extract

The dried egg shells were crushed with mortar and pestle and calcined at 650 °C for 8 h. The calcination process liberated CO₂ with the formation of pure CaO. 0.5 g of calcined CaO was dispersed in 50 mL of distilled water upon constant stirring for 1 h. About 10 mL of tea decoction was added dropwise to the solution and stirred for 1 h. Finally, the solution has to be centrifuged to eliminate the suspended particles and was rinsed with alcohol: water to discard the contaminants. Air dried sample of the prepared material is hence termed as Bio-Nano Calcium oxide (Bio-NaCaO).

2.2. Optimization of microwave assisted chicken feather meal methyl ester (CFMME) or biodiesel synthesis

The optimization study was performed in a 100 mL borosilicate vessel placed in containing 50 mL of CFMO placed in microwave reactor (Fig. 1). The operational temperature was monitored by an IR sensor. The reaction was carried out by varying the temperature from 40 °C to 60 °C with catalysts weight varying from 2 g to 5 g dissolved in varying amount of methanol. Duration of the reaction was also studied between

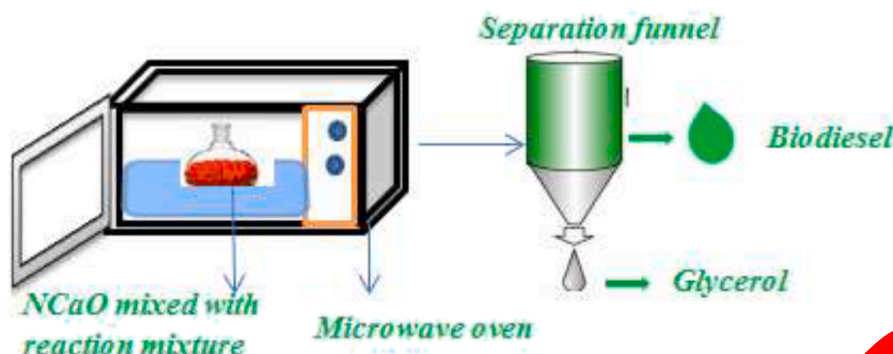


Fig. 1. Reaction setup.

1 min to 5 min at the step speed of 1 min. Complete separation of CFMME, glycerol and catalyst was done using separation funnel left undisturbed for 24 h. Buchirotory evaporator operated at 50 °C was employed to evaporate methanol from prepared CFMME. The obtained CFMME was rinsed with water and heated at 70 °C - 80 °C to evaporate the aqueous content present in it (Fig. 1).

2.3. Characterization of catalysts and analysis of biodiesel

The crystallinity and structure formation was confirmed by PANalytical Xpert PRO XRD diffractometer equipped with copper source at a wavelength (λ) of 1.54060 Å. Manti Lab MT-137 A; a double beam UV-Visible spectrophotometer analysed the absorbance range of synthesized nano particle. The thermal stability of Nano CaO was tested by METTLER TOLEDO TGA/DSC. The morphology of catalyst was studied by JSM-7900F Scanning electron microscope. Gas chromatography analysis of CFMME was done in 7890 series GC set up operated at a temperature of 250 °C with helium flow for 1 mL/min. HP-INNOW Column, 30 m × 0.25 mm, 0.25 µm was inbuilt and an ionization detector (FID) was used to analyse the products. Splitwise temperature moderation was held from 60 °C for 2 min, 10 °C/min to 200 °C, 5 °C/min to 240 °C and 240 °C was held for 7 min. The standard compounds in GC before analysis of CFMME was methyl hexanoate (C6:0) and methyl nervonate (C24:1) dissolved in hexane. The equations used for calculating % yield and % purity of CFMME are given in equation (1).

$$\% \text{ Yield of CFMME} = \frac{\text{Weight of Fatty acid ester}}{\text{Weight of CFMO}} \times 100 \quad (1)$$

2.4. ASTM analysis of CFMME

ASTM standards followed and detailed equipment used to analyse CFMME is given in Table 1. Fatty acid composition of chicken feather meal oil (CFMFO) was analysed by GC provided in Table 2.

3. Results and Discussion

3.1. Characterization of Bio-Nano Calcium oxide (Bio-NCaO)

The XRD result shown in Fig. 2 clearly indicates the formation of expected Bio-NCaO. Four significant peaks were observed at 2θ value of 22.945, 29.274, 39.303 and 47.369 which corresponds to [111], [200] and [211] planes of diffraction (Chaudhary et al., 2018), (Badnore et al.,

Table 1
Specifications of ASTM standards and equipment used in analysis of CFMME.

	Viscosity	Heating value	Pour point	Flash point	Cetane number
ASTM standards	ASTM D 445	ASTM D 240	ISS:1448 Part 10	ASTM D 92	ASTM D 976
Equipment employed in the study	Anton Paar'sViscoQC 100/300	C 2000 Automatic Oxygen bomb Calorimeter	Cloud and Pour point apparatus SICBCPPA-01	Abel flash point apparatus	Cetane index analyzer

Table 2
Fatty acid composition of CFMFO and CFMME.

Fatty acid composition	CFMO %	CFMME %
(C16:0) Palmitic acid	29.87	27.86
(C18:2) Linoleic acid	13.02	14.54
(C18:1) Oleic acid	46.12	46.85
(C18:0) Stearic acid	11.04	10.05

2021). XRD reports matched with the commercial nano CaO material. The increasing order of the planes are [111], [211] and [200] according to the intensity of the peaks. The interplanar distance calculated at the significant peak values were 3.87291 Å, 2.29050 Å, 2.22050 Å and 1.91760 Å. Debye Scherer equation was applied to calculate particle size and the mean particle size was calculated to be 14.1 nm.

$$D = \frac{K\lambda}{\beta \cos \theta} \quad (2)$$

3.2. UV-vis spectrum of Bio-NCaO

The absorbance range of Bio-NCaO is shown in Fig. 3. UV radiations irradiated the sample in the range of 200 nm–450 nm and the graph was plotted between wavelength and absorption. The synthesized material showed a broad peak at 450 nm and there was no absorption in visible region which is characteristic of nano materials (Khanzode et al., 2021; Sree et al., 2020).

3.3. FTIR spectrum of Bio-NCaO

FTIR spectrum of synthesized Bio-NCaO showed characteristic peaks of O–H bond, N–H bond and C–O bond stretching in Fig. 4. A peak around 2929 cm⁻¹ corresponded to O–H stretch. A high resolution peak was observed at 2923 cm⁻¹ indicated the presence of amines in the as-synthesized catalyst. Egg shells used as precursors for the synthesis and the presence of proteins in the form of amides should have contributed to the corresponding peak (Fouad et al., 2020; Khalifeh and Esmaeili, 2020). Peaks at 1444.42 cm⁻¹, 1064.5 cm⁻¹ and 863.9 cm⁻¹ indicated C–O bond of Bio-NCaO. The complete transformation of CaCO₃; the basic constituent of egg shell is confirmed by the absence of peak at 1415 cm⁻¹.

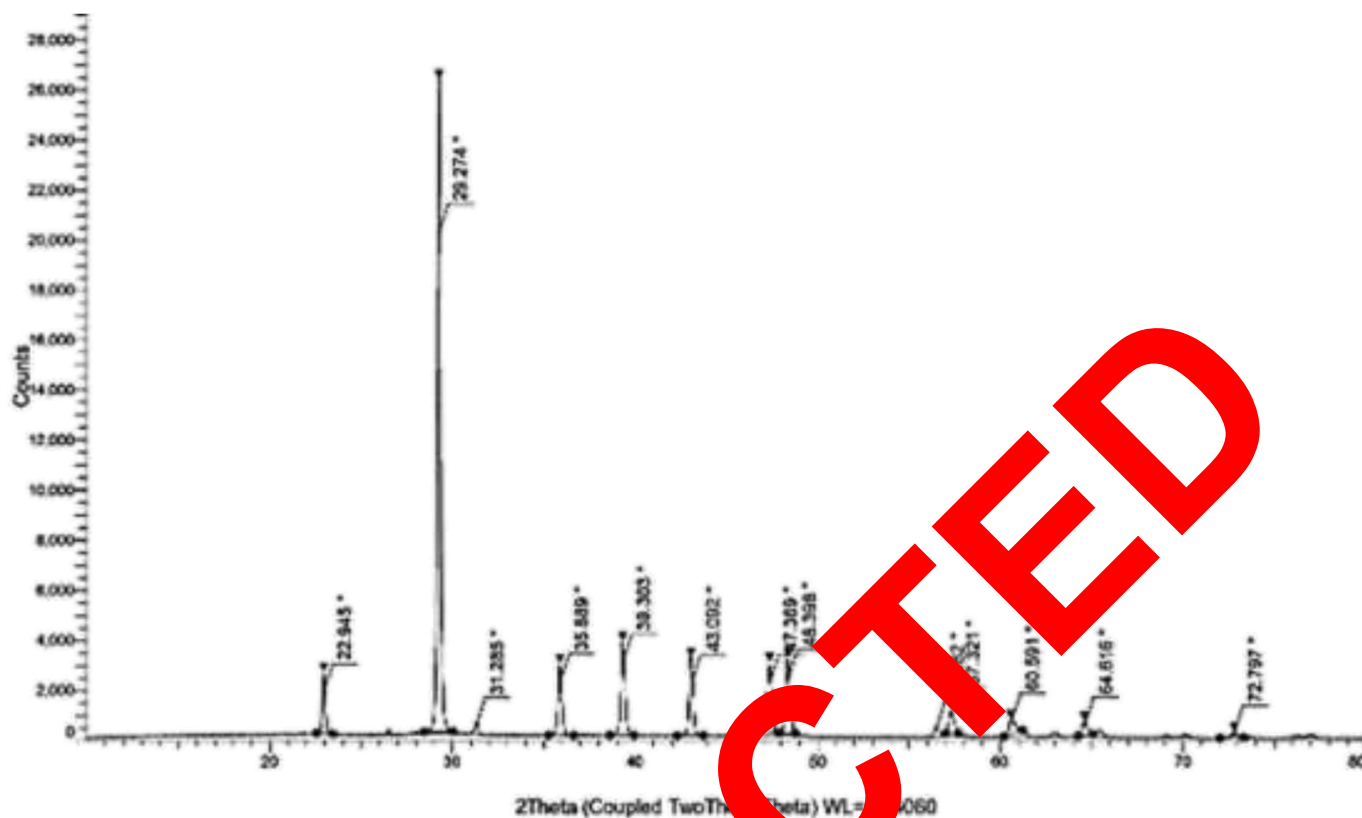


Fig. 2. XRD pattern of Bio-NCaO.

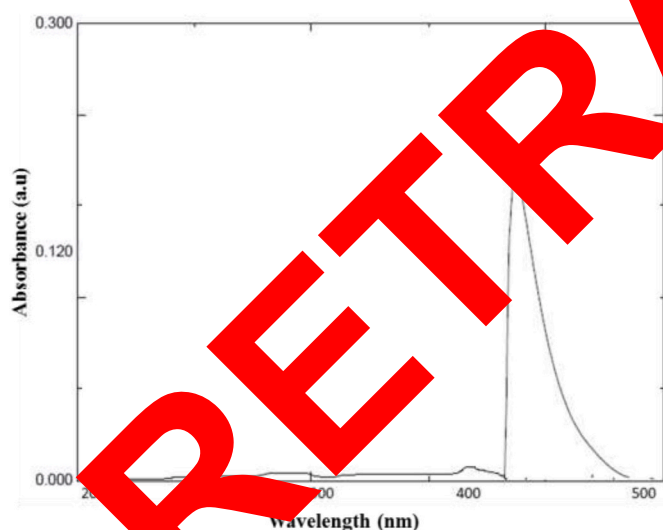


Fig. 3. UV-Vis spectra of Bio-NCaO.

3.4. TGA study of Bio-NCaO

Fig. 5 shows the Thermal stability nature of NCaO catalyst as a function of temperature. The material showed weight loss as calculated from TGA curve. The catalyst showed decomposition in the region of 461°C to 691 °C which can be related to conversion of CaCO_3 to CaO and CO_2 . After the synthesis of nanoparticle, lime was found to be the major component (85.83%).

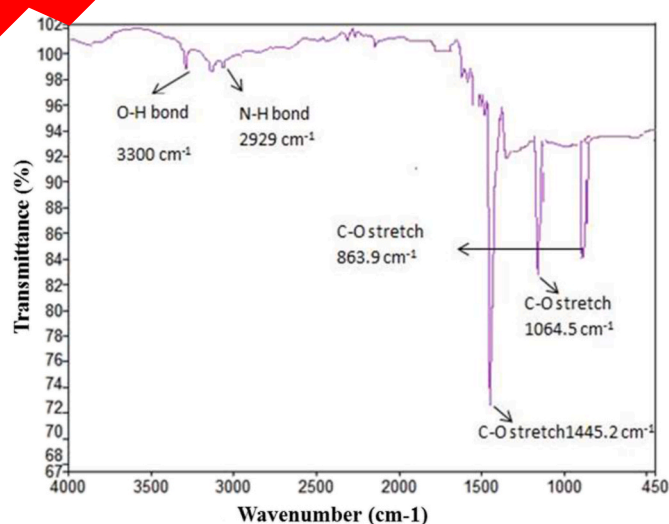


Fig. 4. FTIR spectrum of Bio-NCaO.

3.5. Scanning electron microscope image of Bio-NCaO

SEM image of Bio-NCaO is shown in Fig. 6. Morphology of the catalyst was captured in the 100 μm and 5 μm magnifications. The magnified image of catalyst at 100 μm shown in Fig. 6a displayed discrete and non-agglomerated particles. The same material in the magnification of 5 μm , looked oval and cubic in shape and of different sizes (Fig. 6b). Similar morphology was reported by Wahyuningsih et al. (2019), Habte et al. (2019) and Keihani et al. (2018). SEM image of unmodified egg shell is shown in Fig. 6c for comparison purpose. The non-porous and irregular shape of egg shell is clearly seen from the

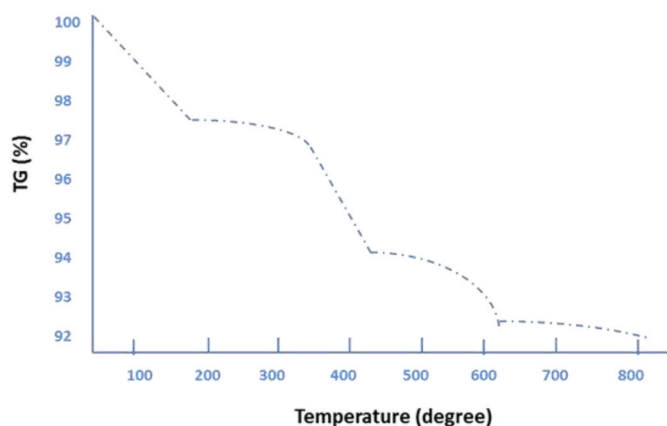


Fig. 5. TGA study of Bio-NCaO.

picture and this proves the egg shell is modified.

3.6. CFMME production using Bio-NCaO

3.6.1. Effect of microwave power

Efficiency of the reaction can be revised with the effect of optimum power of microwave radiation. The CFMME production was studied by varying the microwave radiation power from 200 W to 800 W at methanol: oil ratio of 8, 5 min and 1% catalyst concentration. Fig. 7a depicts the dependence of product yield on the power irradiated on the reaction mixture. The steady increase in power from 200 W to 500 W directly influenced the product formation and further increase in power to 800 W witnessed a significant drop in CFMME yield. The yield of products ranged from 78% to 70% and the highest yield of 95% was

achieved at 500 W. A minimized heat loss induced by the microwave has assisted in accelerated and efficient Transesterification; however, the rise in temperature above a optimum level could lead to more FFAs formed by hydrolysis of triacylglycerols. The excess heat could also evaporate methanol from the mixture which disturbs the reaction of CFMO and alcohol. Dehghan et al. prepared biodiesel from olive oil under microwave radiation and reported the decrease in product yield while exceeding 500 W (Dehghan et al., 2019). Purandaradas et al. (2018) Converted chicken feather meal oil into 88.5% of Biodiesel using KOH catalyst. 90.2% of Biodiesel yield was obtained from waste chicken fat by using CaO catalysts prepared from chicken egg shell (Odetoeye et al., 2021). Whereas, the conversion of CFMO under microwave irradiation was found to be highly favourable.

3.6.2. Effect of methanol: CFMO ratio

The solvent is responsible for complete solubility of the catalyst and uniform dispersion of microwave radiation in the medium. The better activity of Methanol should be because of its high dielectric tangent. Fig. 7b shows the effect of methanol on Transesterification reaction of CFMME in the presence of Bio-NCaO. The molar ratio of Methanol: CFMME was studied from 2:1 to 8:1 and 75% yield were similar to the expected results with the increase in molar ratio from 2:1 to 8:1, Biodiesel yield increased from 75% to 85%. The initial low yield of Biodiesel must be due to the insufficient solvent medium to transfer the energy in reaction medium. However, with further increase in Methanol concentration in the reaction mixture the % yield dropped to less than 8%. The observation could be justified due to a) low mass transfer between the phases b) dilution to catalyst c) reverse reaction and d) increased CFMME in glycerol. Yongphet et al. (2021) reported effective conversion of waste oil into biodiesel over 5 Kw electric power at 6:1 methanol: oil ratio. Degfie et al. (2019) optimized methanol: oil ratio as 1:1 for producing biodiesel from waste cooking oil using Nano CaO

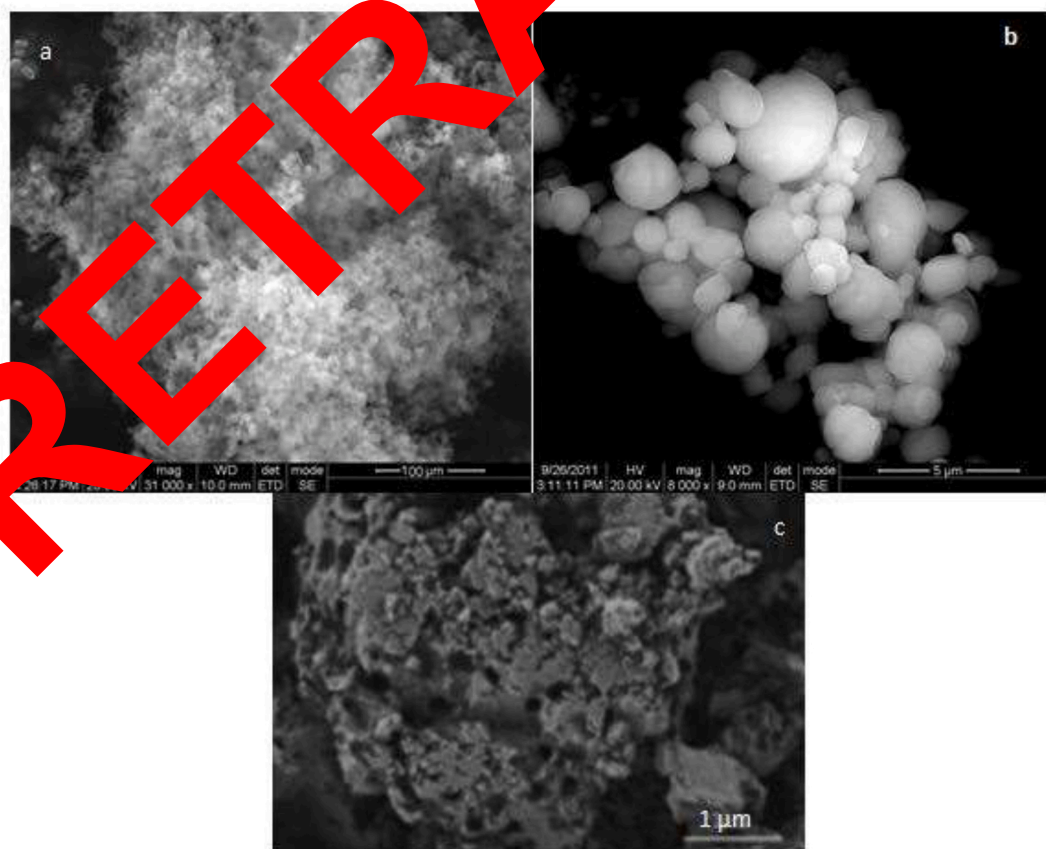


Fig. 6. SEM image of Bio-NCaO a) at 100 μm b) 5 μm.

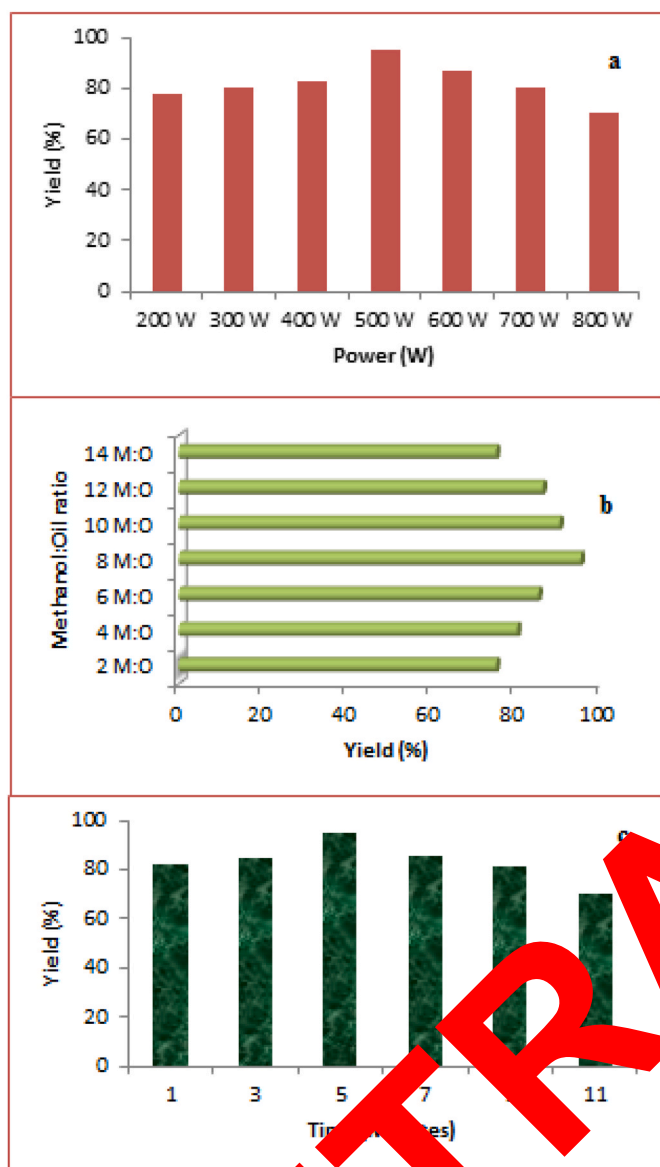


Fig. 7. Optimization of reaction parameters a) Effect of microwave power b) Effect of methanol: CFMO ratio and c) Effect of reaction time.

catalyst.

3.6.3. Effect of reaction time

The time required for completion of reaction with maximum product yield was optimized in Fig. 7c. The reaction was carried out at 500 W, 8:1 methanol:oil ratio and 1% catalyst concentration. The process was performed by varying the time from 1 to 11 min and duration of reaction was observed to affect the reaction. The CFMME yield increased till 5 min and dropped with further increase. The lowest yield of 74% was obtained at first minute and highest yield of 95% at 5 min. A sudden change in yield was noticed on increasing the duration of reaction ahead. The yield reduced from 95% to 86%, 81% and finally to 70%. The reaction efficiency widely depends on the operational temperature as it can afford the required encounter between radiation and reaction mixture. The extended duration will result in solvent evaporation due to the high power of radiation. The reaction is reversible and it may produce unnecessary side products with increase in reaction time. Sharma et al. (2019) converted cotton seed oil into biodiesel and reported the optimized time of reaction as 9.67 min.

3.6.4. Effect of catalyst concentration

Catalyst which accelerates the reaction is important and its concentration plays a crucial role in the conversion. The effect of catalyst on the % yield of CFMME is given in Table 3. The concentration was varied from 0.2% to 1.6% and least conversion of 75% was attained at 0.2%. The shortage of catalyst surface to deliver the products may be the prominent reason for the observation. A gradual increase in yield with increase in concentration was noted till 1% concentration of catalyst. Increasing the concentration above 1% decreased the CFMME yield as contributed to saponification reaction. Fatty acid gets solubilized in soap and therefore hinders the CFMME separation making the reaction procedure tedious. The viscosity of medium might increase with increase in catalyst concentration and affect the separation of glycerol. The presence of excess catalyst also makes catalyst recovery process questionable. Response Surface Methodology (RSM) was used in optimizing synthesis of Fatty Acid Methyl Ester (FAME) from waste cooking oil by Chumuang and Punsuvon (2020). Approximately 99% biodiesel was reported at 2.83 wt% of calcium methoxide catalyst. Numerous articles have supported the current work findings Bharti et al. (2019) and Fayyazi et al. (2018).

3.7. ASTM and properties of CFMME

ASTM standard of petrodiesel is compared with the Biodiesel properties of CFMME in Table 4. In addition, properties of Biodiesel prepared from waste oil prepared by Anu et al. (2018) and from Used cooking oil (UCO) using CaO by Gopan et al. (2021) and Kaewdaeng et al. (2017) by transesterification using Nano CaO catalyst is added to the Table. The calorific value of the fuel determines the calibre of that fuel. Heating value of Used cooking oil Biodiesel (UCB; 42.43 MJ/kg) and Neem biodiesel (NB; 43 MJ/kg) was reported to be less than CFMME (50 MJ/kg) produced in the work. The higher flash point reduces the storage and transportation issues and CFMME showed 153 °C which is lower than UCB and NB Flash points. Trans-esterification reaction has efficiently converted the long chain hydrocarbons into diesel (C18) range which reduced the viscosity of the fuel. Viscosity of CFMME (4.15 mm²/s) was fairly low than NB (5.81 mm²/s) and UCB (4.69 mm²/s). Pour point is the temperature at which the liquid medium loses the mobility and the engines get clogged. The performance of fuel in cold conditions is very important for employment in defence areas of India located in cold regions. Pour point of CFMME is observed to be in the range of NB and UCB. Modifications can be achieved by varying blending ratios of CFMME with petrodiesel. The efficiency of Biodiesel is measured by its cetane number and CFMME has high cetane index (50 min) and this assures the usage of this fuel in near future.

4. Conclusions

It is good to conclude that the synthesized Nano Calcium oxide was absolutely chemicals free and the methodology is green and environmentally benign. In addition, energy efficient and less time consuming microwave irradiation was practised in generating methyl ester. The optimized operational conditions to convert 95% CFMO into Biodiesel were identified to be 500 W, 5 min of reaction, 8:1 methanol: CFMO and 1% catalyst concentration. Comparatively higher heating value of 50 MJ/kg of prepared Biodiesel was in good agreement with ASTM standards of biodiesel. As a result of the research, it can be concluded that low-cost chicken fat could be used as a feedstock and chicken eggshell could be used as a biocatalyst for the effective manufacture of biodiesel. Our future work will focus on investigating the performance of the produced biodiesel in terms of emissions, examining the activity of the catalyst in cracking reactions to produce lighter hydrocarbons, and incorporating acidity to the catalyst to improve the product selectivity.

Table 3

Effect of catalyst concentration on CFMME yield.

Catalyst concentration	0.20%	0.40%	0.60%	0.80%	1.00%	1.20%	1.40%	1.60%
Yield (%)	75	77	82	85	95	90	81	74

Table 4

Comparison of characteristics Calcium oxide catalysed CFMME vs reported biodiesel.

Biodiesel	Heating value (MJ/Kg)	Flash point (°C)	Pour point (°C)	Viscosity mm ² /s	Cetane number
ASTM standard of Biodiesel	46	130	NA	1.9–6.0	47 min
CFMME	50	153	12	4.15	50 min
Used cooking oil biodiesel (UCB)	42.43	157	10	4.69	48 min
Neem biodiesel (Blend30) (NB)	43	175	11	5.81	39.6 min

Credit author statement

Minglong Zhang: Conceptualization, Methodology, Writing- Original draft preparation. **Ganesan Ramya:** Conceptualization, Methodology, Writing- Original draft preparation. **Kathirvel Brindhadevi:** Writing-Reviewing and Editing. **Mishal Alsehli:** Research Support and resources. **Ashraf Elfasakhany:** Writing review and Funding acquisition. **Changlei Xia:** Supervision, Funding acquisition, and Writing-Reviewing and Editing. **Nguyen Thuy Lan Chi:** Reviewing and Editing. **Arivalagan Pugazhendhi:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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